

A Theory of (Linear-Time) Timed Monitors

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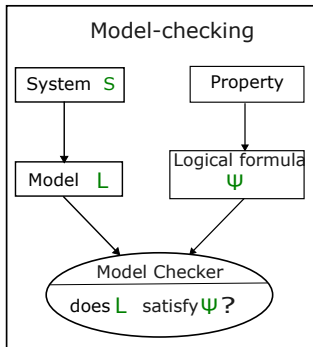
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Formal verification of real-time systems

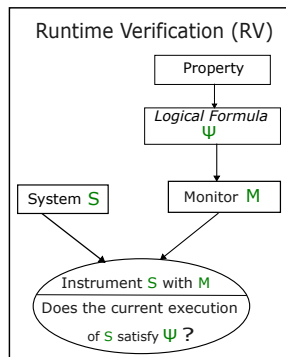
- ▶ In a real-time system, the correctness depends on the order of the actions and the delays that separates them.
- ▶ Certain tasks must be completed within strict time bounds. Violating these constraints can lead to catastrophic consequences.
 - ▶ For example, an autonomous car failing to brake in time to allow a pedestrian to cross.
- ▶ How to ensure that a real-time system behaves as expected ?



Model-checking vs. Runtime Verification



- ▶ Exhaustive verification.
- ▶ A model of the system is needed.
- ▶ State-space explosion.



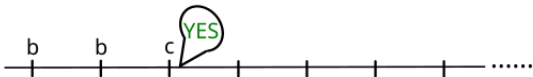
- ▶ Suitable for black-box systems.
- ▶ Scalable.
- ▶ Non-exhaustive.
- ▶ Monitorability.

Monitorability

Some properties are *not monitorable* (on infinite executions).

Example (The 'leads to' property)

Every action **b** is eventually followed by (leads to) an action **c**.

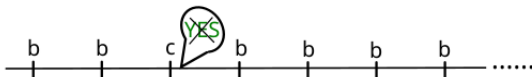
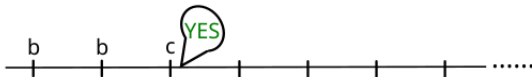


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Monitorability in a timed setting

We consider both the sequence of actions and the delays between them.

Example (The 'bounded response' property)

Every action **b** is eventually followed by an action **c** within 1 time unit (relative to the occurrence of **b**).

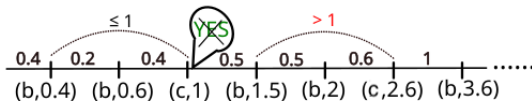


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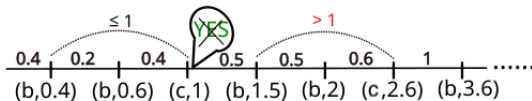


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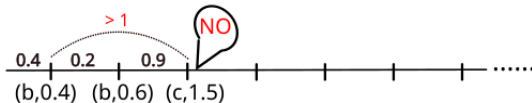
This property is not monitorable for satisfactions!

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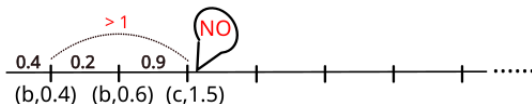


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Example (The 'bounded response' property)

Every action **b** is eventually followed by an action **c** within 1 time unit (relative to the occurrence of **b**).



This property is monitorable for violations...

The Monitorability problem in a timed setting

- ▶ In a nutshell, the “game” of monitorability is about telling the future.
- ▶ Given some formula ϕ , can I correctly issue, after a finite observation, a :
 - ▶ *Correct* satisfaction verdict for every infinite execution satisfying ϕ (satisfaction monitorability),
 - ▶ *Correct* violation verdict for every infinite execution violating ϕ (violation monitorability).

Problem

Given an expressive Timed Logic, which formulae are monitorable?

The monitorability problem has been solved **only** for *untimed* logics, notably the untimed modal μ -calculus.

Real-time executions

- ▶ Given a finite set of actions ACT , a silent action τ , we define the set of labels $LAB = ACT \cup \{\tau\} \cup \mathbb{R}_{\geq 0}$.
- ▶ A real-time system is a timed labelled transition system (TLTS) over LAB .
- ▶ A real-time execution (run) starting from a program (state) p_1 is an infinite sequence of transitions in the TLTS.

▶ for example

$$p_1 \xrightarrow{0.5} p_2 \xrightarrow{\tau} p_3 \xrightarrow{0.3} p_4 \xrightarrow{b} p_5 \xrightarrow{1.2} p_6 \xrightarrow{c} p_7 \dots$$

- ▶ $0.5 \tau 0.3 b 1.2 c \dots$ is a run (in LAB^ω) starting from p_1 .
- ▶ Let $\mathcal{L} = LAB \setminus \{\tau\}$. A finite weak (observable) run in $\mathcal{L}^*$ is obtained from a prefix of a run
 - ▶ example $0.8 b 1.2 c$ from $0.5 \tau 0.3 b 1.2 c$
- ▶ for some states p, p' and weak finite run $\theta \in \mathcal{L}^*$, we write:
 - ▶ $p \xrightarrow{\theta} p'$ to denote that p' is reachable from p along θ ,
 - ▶ $p \xrightarrow{\theta}_{inev} p'$ to denote that p' is the only reachable state from p along θ .

Real-time executions

- ▶ An *infinite-timed-word* (trace) representation is used to evaluate logical formulae.
- ▶ Let $\text{TACT} = \text{ACT} \times \mathbb{R}_{\geq 0}$. A trace is of the form $(a_1, t_1)(a_2, t_2) \dots$ in TACT^ω such that $t_i \leq t_{i+1}$ for every $i \in \mathbb{N}_{>0}$.
 - ▶ Here the *timestamp* t_i represents the *absolute time* at which action a_i occurred.
 - ▶ A finite trace in TACT^* is a prefix of a trace.
 - ▶ Each finite trace *corresponds* therefore to a unique finite weak run using $\text{TRN} : \text{TACT}^* \longrightarrow \mathcal{L}^*$:

$$\text{TRN}((a_1, t_1)(a_2, t_2) \dots (a_n, t_n)) = d_1 \ a_1 \ d_2 \ a_2 \dots \ d_n \ a_n$$

$$\text{with } d_i = t_i - t_{i-1} \text{ and } t_0 = 0$$

- ▶ Example: $\text{TRN}((a, 1.7)(b, 1.7)(b, 5.7)) = 1.7 \ a \ 0 \ b \ 4 \ b$

Intervals

Definition (Interval syntax)

The set $\mathcal{I}$ is given by the grammar $I \in \mathcal{I} ::= \sqsubset p, q \sqsupset$ where $\sqsubset \in \{[, (\}$, $\sqsupset \in \{],) \}$, $p \in \mathbb{N}$ and $q \in \mathbb{N} \cup \{\infty\}$. We require that $\sqsupset =)$ if $q = \infty$.

- ▶ p (resp. q) is called the left (resp. right) bound of I , we write accordingly $\mathcal{LB}(I) = p$ (resp. $\mathcal{RB}(I) = q$).
- ▶ I is said *bounded* if $\mathcal{RB}(I) \neq \infty$.

Definition (Interval Semantics)

$$\llbracket I \rrbracket = \{x \in \mathbb{R}_{\geq 0} \mid p \prec_I x \prec_r q\}$$

where

- ▶ $\prec_I = \leq$ if $\sqsubset = [$ and $\prec_I = <$ otherwise,
- ▶ $\prec_r = \leq$ if $\sqsupset =]$ and $\prec_r = <$ otherwise.
- ▶ We consider only non-empty intervals: $\{I \in \mathcal{I} \mid \llbracket I \rrbracket \neq \emptyset\}$,
- ▶ We use I to denote both the syntax and semantics of an interval I .

MTL

Metric Temporal Logic (MTL)

$$\phi, \psi ::= a \mid \neg a \mid \phi \wedge \psi \mid \phi \vee \psi \mid \bigcirc_I \phi \mid \phi \mathcal{U}_I \psi \mid G_I \phi$$

$\llbracket \mathbf{b} \mathcal{U}_I \mathbf{c} \rrbracket$ contains all infinite traces where only **b** actions occur **until** **c** occurs **within** interval I (relative to the first timestamp in the trace).

$\pi \in \llbracket \mathbf{b} \mathcal{U}_I \mathbf{c} \rrbracket$ iff $\pi \in \llbracket \mathbf{b} \mathcal{U} \mathbf{c} \rrbracket$ **and** the first occurrence of **c** is within I .

Example

Let $\pi = (\mathbf{b}, 3)(\mathbf{a}, 4)(\mathbf{c}, 7) \dots$, $\pi' = (\mathbf{b}, 3)(\mathbf{b}, 4)(\mathbf{c}, 7) \dots$ and $\pi'' = (\mathbf{b}, 2)(\mathbf{c}, 3) \dots$

- ▶ $\pi \notin \llbracket \mathbf{b} \mathcal{U}_{[2,5]} \mathbf{c} \rrbracket$ because $\pi \notin \llbracket \mathbf{b} \mathcal{U} \mathbf{c} \rrbracket$.
- ▶ $\pi' \in \llbracket \mathbf{b} \mathcal{U}_{[2,5]} \mathbf{c} \rrbracket$ because $\pi' \in \llbracket \mathbf{b} \mathcal{U} \mathbf{c} \rrbracket$ and $7 - 3 \in [2, 5]$.
- ▶ $\pi'' \notin \llbracket \mathbf{b} \mathcal{U}_{[2,5]} \mathbf{c} \rrbracket$ because **c** occurs too early ($3 - 2 \notin [2, 5]$).

MTL: Limitations

- ▶ Hard to reason about formulae (e.g., using induction).
- ▶ Limited expressivity.
 - ▶ MTL cannot express properties that involve *counting*.
 - ▶ Example: the following property is not expressible in MTL:
For every even position in the trace, if the action happens within the interval I , then the action is **b**.

Our timed mu-calculus T^{lin}

$\phi, \psi \in T^{lin}$	$::=$	tt	(true)
		ff	(false)
		X	(logical variable)
		$\phi \wedge \psi$	(conjunction)
		$\phi \vee \psi$	(disjunction)
		$\langle a \rangle \phi$	(diamond)
		$\langle g \rangle \phi$	(diamond on time)
		$x \text{ in } \phi$	(timestamp recording)
		$\min(X, \phi)$	(min. fixed-point)
		$\max(X, \phi)$	(max. fixed-point)

$$g ::= x \prec n \quad | \quad (x - y) \prec n \quad | \quad n \prec x \quad | \quad n \prec (x - y) \quad | \quad g \wedge g \quad | \quad \neg g$$

- ▶ X is a **recursion variable**, x, y are **timestamp variables**, n is an integer.
- ▶ A **formula** is a **closed** term generated by the grammar above.

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- ▶ Notation shortcuts: given non empty $A \subseteq \text{ACT}$, $\langle A \rangle \phi = \bigvee_{a \in A} \langle a \rangle \phi$, $\bar{a} = \text{ACT} \setminus \{a\}$, $\bar{g} = \neg g$.
- ▶ syntactic sugar: $[g]\phi = \langle g \rangle \phi \vee \langle \bar{g} \rangle \text{tt}$, $[a]\phi = \langle a \rangle \phi \vee \langle \bar{a} \rangle \text{tt}$.

Semantics of T^{lin} : a (simple) example

Example

The property 'The first action is *start* and occurs within 2 time units' is expressed by the T^{lin} formula:

$$x \text{ \underline{in} } \langle x \leq 3 \rangle \langle start \rangle \text{tt}$$

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$$\pi = (b, 3.4) \dots$$

Semantics of T^{lin} : a (not-so-simple) example

Let $\pi \in \text{TACT}^\omega$ be a trace and $\phi_{\text{even}(b), I} \in T^{lin}$ a formula defined as follows

$$x \text{ \underline{in} } \overbrace{\max(X, \langle \text{ACT} \rangle (y \text{ \underline{in} } \underbrace{\langle y - x \in I \rangle \langle \mathbf{b} \rangle X}_{\psi_1}) \vee (z \text{ \underline{in} } \underbrace{\langle z - x \notin I \rangle \langle \text{ACT} \rangle X \rangle}_{\psi_2}))}_{\psi}$$

- ▶ $x \text{ \underline{in} }$ records the first timestamp of π .
- ▶ π is in $\llbracket \phi_{\text{even}(b), I} \rrbracket$ if the first action is any action and the continuation of π (π^2) is in $\llbracket \psi_1 \rrbracket$ or $\llbracket \psi_2 \rrbracket$.
- ▶ ψ_1 checks that the second action of π is a $\mathbf{b}$ and occurs within I , and the continuation of π^2 (π^3) is in $\llbracket \psi \rrbracket$.
- ▶ ψ_2 checks that the second action of π does not occur within I and π^3 is in $\llbracket \psi \rrbracket$.

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- ▶ ψ_2 checks that the second action of π does not occur within I and π^3 is in $\llbracket \psi \rrbracket$.
- ▶ Since we use a maximal fixed point, π is in $\llbracket \phi_{\text{even}(b),I} \rrbracket$ iff at every even position of π , if the action occurs within I then it is a **b**.

Semantics of T^{lin} : a (not-so-simple) example

Let $\phi_{even(b),l} \in T^{lin}$ a formula defined as follows

$$x \text{ \underline{in} } \overbrace{\max(X, \langle \text{ACT} \rangle (y \text{ \underline{in} } \underbrace{\langle y - x \in l \rangle \langle \mathbf{b} \rangle X}_{\psi_1} \vee (z \text{ \underline{in} } \underbrace{\langle z - x \notin l \rangle \langle \text{ACT} \rangle X)_{\psi_2}))}_{\psi}$$

- ▶ π is in $\llbracket \phi_{even(b),l} \rrbracket$ iff at every even position of π , if the action occurs within l then it is a **b**.
- ▶ example for $\pi = (c, 1)(b, 2.5)(c, 8) \dots$ is in $\llbracket \phi_{even(b),[2,6]} \rrbracket$,
- ▶ $\pi = (c, 1)(b, 3.5)(b, 5)(c, 8) \dots$ is in $\llbracket \phi_{even(b),[2,6]} \rrbracket$,
- ▶ $\pi = (c, 0)(b, 3.5)(b, 5)(c, 5.5)(b, 8) \dots$ is **not** in $\llbracket \phi_{even(b),[2,6]} \rrbracket$.

Expressiveness

Theorem

T^{lin} is strictly more expressive than MTL:

1. For every $\phi \in MTL$, there exists $\psi \in T^{lin}$ such that $\llbracket \phi \rrbracket = \llbracket \psi \rrbracket$
2. There exists $\psi \in T^{lin}$ such that for every $\phi \in MTL$ we have $\llbracket \phi \rrbracket \neq \llbracket \psi \rrbracket$

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1. For every $\phi \in MTL$, there exists $\psi \in T^{lin}$ such that $\llbracket \phi \rrbracket = \llbracket \psi \rrbracket$
- We translate MTL formulae to equivalent T^{lin} formulae via $TO T^{lin} : MTL \longrightarrow T^{lin}$ such that:
- For every $\phi \in MTL$ we have $\llbracket \phi \rrbracket = \llbracket TO T^{lin}(\phi) \rrbracket$.

Expressiveness

Example

Let $\phi = \mathbf{b} \mathcal{U}_I \mathbf{c}$. We have

$$\text{TO } T^{lin}(\phi) = x_s \underline{\text{in}} \overbrace{\min(X, \underbrace{(x_e \underline{\text{in}} \langle x_e - x_s \in I \rangle \langle \mathbf{c} \rangle \mathbf{tt})}_{\psi_1} \vee \underbrace{(\langle \mathbf{b} \rangle \mathbf{tt} \wedge \langle \text{ACT} \rangle X)}_{\psi_2})}_{\psi}$$

- ▶ We use a minimal fixed point because we need to find a position with action $\mathbf{c}$.
- ▶ ψ_1 checks if the action occurs within I , and it is a $\mathbf{c}$.
- ▶ ψ_2 checks that the action is a $\mathbf{b}$ and the continuation of the trace is in $\llbracket \psi \rrbracket$.
- ▶ Therefore, the semantics of ϕ and $\text{TO } T^{lin}(\phi)$ coincide.

Expressiveness

Theorem

T^{lin} is strictly more expressive than MTL:

2. There exists $\psi \in T^{lin}$ such that for every $\phi \in MTL$ we have $\llbracket \phi \rrbracket \neq \llbracket \psi \rrbracket$

Example: the T^{lin} formula $\phi_{even(b),I}$ has no equivalent counterpart in MTL.

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the following formula ϕ

$$x \text{ \underline{in} } \max(X, y \text{ \underline{in} } (\langle y - x \in I \rangle \langle \mathbf{c} \rangle (z \text{ \underline{in} } (\langle z - x \in I \rangle \langle \mathbf{b} \rangle X \vee \langle z - x \notin I \rangle \langle \text{ACT} \rangle X))) \\ \vee (\langle y - x \notin I \rangle \langle \text{ACT} \rangle \mathbf{tt}))$$

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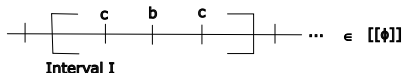
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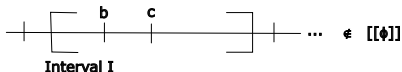
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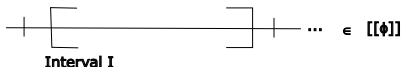
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has no equivalent counterpart in MTL.



Semantic properties

- ▶ $\llbracket \max(X, \phi), \rho \rrbracket_v = \bigcup \{S \mid S \subseteq \llbracket \phi, \rho[X \mapsto S] \rrbracket_v\}_v$
- ▶ $\llbracket \min(X, \phi), \rho \rrbracket_v = \bigcap \{S \mid \llbracket \phi, \rho[X \mapsto S] \rrbracket_v \subseteq S\}_v$

For all term ϕ , environment ρ and valuation v : $\lambda x. \llbracket \phi, \rho[X \mapsto x] \rrbracket_v$ is monotone and continuous.

- ▶ Monotonicity + Knaster-Tarski's fixpoint theorem $\implies$
 - ▶ $\llbracket \max(X, \phi), \rho \rrbracket_v$ is the greatest fixed point of $\lambda x. \llbracket \phi, \rho[X \mapsto x] \rrbracket_v$.
 - ▶ $\llbracket \min(X, \phi), \rho \rrbracket_v$ is the last fixed point of $\lambda x. \llbracket \phi, \rho[X \mapsto x] \rrbracket_v$.
- ▶ Continuity + Kleene fixed point theorem $\implies$ Inductive reasoning on **min.** formulae.

Monitors

$M, N \in \text{MON}$	$::= E$	$a.M$	X
	$g.M$	$M + N$	$\text{rec } X.M$
	$\text{LET } x \text{ IN } M$	$M \otimes N$	$M \oplus N$
E	$::= \text{VYES}$	VNO	VEND
VYES	$::= \text{YES}$	$\text{VYES} \oplus M$	$\text{VYES} \otimes \text{VYES}$
VNO	$::= \text{NO}$	$\text{VNO} \otimes M$	$\text{VNO} \oplus \text{VNO}$
VEND	$::= \text{END}$	$\text{VEND} \oplus \text{VEND}$	$\text{VEND} \otimes \text{VEND}$

Monitor semantics

$\mathbb{M}$ is the TLTS representing the semantics of all monitors in MON .

- ▶ A monitor state is a triplet (M, v_c, v_t) where $M \in \text{MON}$, $v_c \in \mathbb{R}_{\geq 0}$ is a global “clock” denoting the absolute time and v_t is a valuation of timestamp variables.
- ▶ Given M we let $m_0 = (M, 0, \mathbf{0})$ be its initial state.
- ▶ Verdict states equivalence :
 - ▶ M and N are in VYES implies $(M, v_c, v_t) = (N, v'_c, v'_t) = \text{yes}$
 - ▶ M and N are in VNO implies $(M, v_c, v_t) = (N, v'_c, v'_t) = \text{no}$
 - ▶ M and N are in VEND implies $(M, v_c, v_t) = (N, v'_c, v'_t) = \text{end}$
- ▶ Transition rules are defined analogously to the evaluation of T^{lin} formulae (examples to follow).

Instrumentation

A monitor is a *passive entity*, it is *at the mercy* of the monitored program.

$$\begin{array}{c}
 \frac{p \xrightarrow{\alpha}_P p' \quad m \xrightarrow{\alpha}_M m' \quad \alpha \in \text{TACT}}{m \triangleleft p \xrightarrow{\alpha}_{\triangleleft} m' \triangleleft p'} \text{IMON} \qquad \frac{p \xrightarrow{\tau}_P p'}{m \triangleleft p \xrightarrow{\tau}_{\triangleleft} m \triangleleft p'} \text{IASY} \\
 \\
 \frac{p \xrightarrow{a}_P p' \quad m \not\xrightarrow{a}_M \quad m = (N, v_c, v_t)}{m \triangleleft p \xrightarrow{a}_{\triangleleft} (\text{END}, v_c, v_t) \triangleleft p'} \text{ITER}
 \end{array}$$

Instrumentation : a (simple) example

Let

$$M = \text{LET } x \text{ IN } ((x \in [1, 3]). \underbrace{\text{rec } X. (\mathbf{b}.X + \mathbf{c}.\text{NO})}_{N} + (x \notin [1, 3]).\text{YES}))$$

and $p \xrightarrow{1.6} p_1 \xrightarrow{b} p_2 \xrightarrow{c} p_3 \xrightarrow{4} p_4 \xrightarrow{b} p_5 \dots$. Then we have

$$m_0 \triangleleft p \xrightarrow{1.6} (M, 1.6, \mathbf{0}) \triangleleft p_1 \xrightarrow{b} (N, 1.6, [x = 1.6]) \triangleleft p_2 \xrightarrow{c} \text{no} \triangleleft p_3 \xrightarrow{4} \text{no} \triangleleft p_4 \xrightarrow{b} \dots$$

- ▶ Over $\text{ACT} = \{b, c\}$, M intuitively monitors for the formula $x \text{ in}[x \in [1, 3]] \max(X, \langle b \rangle X)$.
- ▶ Informally: If the first action occurs with a timestamp in $[1, 3]$, then every action must be a **b**.

Reactive monitors

What about $\text{ACT} = \{b, c, c'\}$? Let

$$M = \text{LET } x \text{ IN } ((x \in [1, 3]). \underbrace{\text{rec } X. (\mathbf{b}.X + \mathbf{c}.\text{NO})}_{N} + (x \notin [1, 3]). \text{YES}))$$

and $p \xrightarrow{1.6} p_1 \xrightarrow{b} p_2 \xrightarrow{c'} p_3 \xrightarrow{4} p_4 \xrightarrow{b} p_5 \dots$ Then we have

$$m_0 \triangleleft p \xrightarrow{1.6} (M, 1.6, \mathbf{0}) \triangleleft p_1 \xrightarrow{b} (N, 1.6, [x = 1.6]) \triangleleft p_2 \xrightarrow{c'} \text{end} \triangleleft p_3 \xrightarrow{4} \text{end} \triangleleft p_4 \xrightarrow{b} \dots$$

M is now not reactive, it does not know what to say!

Since we are in a linear-time setting, we can:

- ▶ consider only reactive monitors (without loss of generality),
- ▶ abstract away the monitored program into the set of runs it produces.
 - ▶ we write accordingly $m \xrightarrow{\theta} m'$ instead of $m \triangleleft p \xrightarrow{\theta} m' \triangleleft p'$ for some p and p' ,
 - ▶ and reason on all finite weak runs in $\mathcal{L}^*$ (and consequently on all weak runs in $\mathcal{L}^\omega$)

Monitorability: setting

Time-divergent traces:

- ▶ $\pi \in \text{TACT}^\omega$ is said time divergent iff for every positive real r , there exists a position in π with timestamp greater than r .
- ▶ we denote TACT_{div}^ω the set of time-divergent traces.

This assumption is natural, since in reality time always diverges.

- ▶ example: the trace $(a_1, \frac{1}{2})(a_2, \frac{3}{4})(a_3, \frac{7}{8})(a_4, \frac{15}{16}) \dots$ is not realistic (global time bounded by 1),
- ▶ no other assumption is made: actions can happen at the same time and delays may decrease arbitrarily as long as time eventually diverges.
- ▶ exemple $(a_1, 0)(a_2, 0) \dots (a_6, \frac{1}{2})(a_7, \frac{3}{4})(a_8, \frac{7}{8}) \dots (a_{20}, 1)(a_{21}, 2) \dots$ is in TACT_{div}^ω .

Monitorability: setting

We define the set of prefixes of a trace $\pi \in \text{TACT}^\omega$

$$Pf(\pi) = \{\pi' \in \text{TACT}^* \mid \exists \pi'' \in \text{TACT}^\omega : \pi' \cdot \pi'' = \pi\}$$

and infinite extensions of a finite trace $\pi \in \text{TACT}^*$

$$Ext(\pi) = \{\pi' \in \text{TACT}^\omega \mid \pi \in Pf(\pi')\}$$

Definition (Monitor Soundness)

A monitor M is sound for $\phi \in \mathcal{T}^{lin}$ if for every finite trace $\pi \in \text{TACT}^*$:

- ▶ $m_0 \xrightarrow{\text{TRN}(\pi)} \text{yes}$ implies $\forall \pi' \in Ext(\pi) : \pi' \in \llbracket \phi \rrbracket$
- ▶ $m_0 \xrightarrow{\text{TRN}(\pi)} \text{no}$ implies $\forall \pi' \in Ext(\pi) : \pi' \notin \llbracket \phi \rrbracket$

Soundness is not sufficient, e.g., END is sound for every formula.

Monitorability: strong completeness

Definition (Strong completeness)

- M is strongly satisfaction complete for ϕ if for all $\pi \in \text{TACT}_{div}^{\omega}$:

$$\pi \in \llbracket \phi \rrbracket \text{ implies } \exists \pi' \in Pf(\pi) : m_0 \xrightarrow{\text{TRN}(\pi')}_{inev} \text{ yes}$$

- M is strongly violation complete for ϕ if for all $\pi \in \text{TACT}_{div}^{\omega}$:

$$\pi \notin \llbracket \phi \rrbracket \text{ implies } \exists \pi' \in Pf(\pi) : m_0 \xrightarrow{\text{TRN}(\pi')}_{inev} \text{ no}$$

- M is strongly complete for ϕ if it is both strongly violation and strongly satisfaction complete for ϕ .

Monitorability

A formula $\phi \in \mathcal{T}^{lin}$ is complete (resp. violation, satisfaction) monitorable if there exists a monitor that is sound and strongly complete (resp. strongly violation complete, strongly satisfaction complete) for it.

Problem

What is the largest fragment of $\mathcal{T}^{lin}$ such that every formula is complete (resp. violation, satisfaction) monitorable ?

Sound monitor synthesis

$\llbracket \text{ff} \rrbracket$	=	NO
$\llbracket \text{tt} \rrbracket$	=	YES
$\llbracket X \rrbracket$	=	X
$\llbracket \langle a \rangle \phi \rrbracket$	=	$a. \llbracket \phi \rrbracket + \bar{a}. \text{NO}$
$\llbracket \langle g \rangle \phi \rrbracket$	=	$g. \llbracket \phi \rrbracket + \bar{g}. \text{NO}$
$\llbracket x \text{ in } \phi \rrbracket$	=	LET x IN $\llbracket \phi \rrbracket$
$\llbracket \phi \wedge \psi \rrbracket$	=	$\llbracket \phi \rrbracket \otimes \llbracket \psi \rrbracket$
$\llbracket \phi \vee \psi \rrbracket$	=	$\llbracket \phi \rrbracket \oplus \llbracket \psi \rrbracket$
$\llbracket \max(X, \phi) \rrbracket$	=	$\llbracket \min(X, \phi) \rrbracket = \text{rec } X. \llbracket \phi \rrbracket$

- ▶ The function $\llbracket - \rrbracket$ is rather intuitive and straightforward.
- ▶ For example, for $\langle a \rangle \phi$, we want a monitor that monitors for ϕ if the action is a and reaches verdict NO otherwise.
- ▶ For $\phi \wedge \psi$, we want a parallel monitor that reaches verdict YES iff both its submonitors reach verdict YES.
 - ▶ by definition of monitors, $M \otimes N$ is a YES verdict iff both M and N are YES verdicts.

Theorem

$\llbracket \phi \rrbracket$ is reactive, deterministic and sound for ϕ .

Largest complete-monitorable fragment?

In the untimed (linear-time) μ -calculus, the *recursion-free* fragment is the largest complete-monitorable fragment. This result does not hold in the timed setting.

Consider $\phi = \max(X, \langle \mathbf{b} \rangle X)$ meaning 'globally $\mathbf{b}$ '.

- ▶ $\langle \phi \rangle = \text{rec } X.(b.X + \bar{\mathbf{b}}.\text{NO})$ is sound and violation-complete for ϕ .
- ▶ There is no sound and satisfaction-complete monitor for ϕ ...

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- ▶ There is no sound and satisfaction-complete monitor for ϕ ...

Consider now ϕ_c a *time-constrained* version of ϕ , i.e., using a bounded interval $I = [2, 5]$:

$$\phi_c = x \text{ \underline{in} } \max(X, (y \text{ \underline{in} } [y - x \in [2, 5]] \langle \mathbf{b} \rangle \text{tt}) \wedge \langle \text{ACT} \rangle X)$$

Largest complete-monitorable fragment?

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Consider $\phi = \max(X, \langle \mathbf{b} \rangle X)$ meaning ‘globally **b**’.

- ▶ $\langle \phi \rangle = \text{rec } X.(b.X + \bar{\mathbf{b}}.\text{NO})$ is sound and violation-complete for ϕ .
- ▶ There is no sound and satisfaction-complete monitor for ϕ ...

Consider now ϕ_c a *time-constrained* version of ϕ , i.e., using a bounded interval $I = [2, 5]$:

$$\phi_c = x \text{ \underline{in} } \max(X, (y \text{ \underline{in} } [y - x \in [2, 5]] \langle \mathbf{b} \rangle \text{tt}) \wedge \langle \text{ACT} \rangle X)$$

Meaning “within the interval $[2, 5]$ (relative to the timestamp of the first action), every action must be a **b**”.

Here, a monitor can safely say YES when the time elapsed since the first action exceeds 5.

ϕ_c is therefore complete-monitorable.

Another example: constrained bounded response

- Formula ψ_c below expresses the *constrained* bounded response property, where I and I' are bounded intervals:
 “every action **b** occurring within interval I (relative to the first action) is followed by a **c** occurrence within I' (relative to such occurrence of **b**)”

$$\psi_c = x \underline{\text{in}} \max(Y, (y \underline{\text{in}} [y - x \in I]([b]\text{ff} \vee \zeta)) \wedge (\langle \text{ACT} \rangle Y))$$

with:

$$\zeta = x' \underline{\text{in}} \min(X, (y' \underline{\text{in}} \langle y' - x' \in I' \rangle \langle c \rangle \text{tt}) \vee (\langle \text{ACT} \rangle X))$$

ψ_c is complete monitorable.

Trick to capture the largest complete-monitorable fragment

$$\psi_c = x \text{ \underline{in} } \max(Y, (y \text{ \underline{in} } [y - x \in I]([b]\text{ff} \vee \zeta)) \wedge (z \text{ \underline{in} } [z - x \prec_r \mathcal{RB}(I)]\langle \text{ACT} \rangle Y))$$

with:

$$\zeta = x' \text{ \underline{in} } \min(X, (y' \text{ \underline{in} } \langle y' - x' \in I' \rangle \langle c \rangle \text{tt}) \vee (z' \text{ \underline{in} } \langle z' - x' \prec_r \mathcal{RB}(I') \rangle \langle \text{ACT} \rangle X))$$

- We capture the complete-monitorable formulae by constraining recursion variables in time.

The complete-monitorable fragment of T^{lin}

$$\begin{array}{llll}
 \phi, \psi & ::= \text{tt} & | \text{ff} & | \phi \vee \psi & | \phi \wedge \psi \\
 & | x \underline{\text{in}} \phi & | \langle a \rangle \phi & | \langle g \rangle \phi & \\
 & | \max(X, \phi) & | \min(X, \phi) & | x \underline{\text{in}} \langle g_x \rangle \langle a \rangle X & \\
 g_x & ::= & x \prec n & | x - y \prec n & | g_x \wedge g
 \end{array}$$

Definition (CT^{lin})

$\phi \in CT^{lin}$ iff ϕ is a formula generated by the above grammar, and for every $\max(X, \psi)$, $\min(X, \psi)$ subterms of ϕ , **y is free in ψ** .

Lemma

$\lambda x. \llbracket \phi, \rho[X \mapsto x] \rrbracket_v$ has a unique fixpoint, i.e.,

- ▶ For all $\max(X, \phi) \in CT^{lin}$, $\llbracket \max(X, \phi) \rrbracket = \llbracket \min(X, \phi) \rrbracket$
- ▶ For all $\min(X, \phi) \in CT^{lin}$, $\llbracket \min(X, \phi) \rrbracket = \llbracket \max(X, \phi) \rrbracket$

Theorem

For all $\phi \in CT^{lin}$, $\llbracket \phi \rrbracket$ is sound and strongly complete for ϕ .

Another example: constrained $\phi_{\text{even}(b),I}$

Recall $\phi_{\text{even}(b),[3,7]} \in \mathcal{T}^{\text{lin}}$, not expressible in MTL:

$$x \text{ \underline{in} } \max(X, \langle \text{ACT} \rangle (y \text{ \underline{in} } \langle y - x \in [3, 7] \rangle \langle \mathbf{b} \rangle X) \vee (z \text{ \underline{in} } \langle z - x \notin [3, 7] \rangle \langle \text{ACT} \rangle X))$$

$\phi_{\text{even}(b),[3,7]}$ is also complete monitorable and can be written in $\mathcal{CT}^{\text{lin}}$:

$$x \text{ \underline{in} } \max(X, \langle \text{ACT} \rangle (y \text{ \underline{in} } \langle y - x \in [3, 7] \rangle \langle \mathbf{b} \rangle X) \vee (z \text{ \underline{in} } (\langle z - x < 3 \rangle \langle \text{ACT} \rangle X \vee \langle z - x \geq 7 \rangle \mathbf{tt})))$$

VT^{lin} : the violation-monitorable fragment of T^{lin}

- ▶ Intuition : “relax” the syntax of the complete-monitorable fragment to allow general **max** formulae.

$$\begin{array}{lcl}
 \phi, \psi & ::= & \text{tt} \quad | \quad \text{ff} \quad | \quad \phi \vee \psi \quad | \quad \phi \wedge \psi \\
 & & | \quad x \text{ in } \phi \quad | \quad \langle a \rangle \phi \quad | \quad \langle g \rangle \phi \\
 & & | \quad \max(X, \phi) \quad | \quad X \\
 & & | \quad \cancel{\min(X, \phi)} \quad | \quad \cancel{x \text{ in } \langle g_x \rangle \langle a \rangle X}
 \end{array}$$

- ▶ example: $\phi_{\text{even}(b), [0, \infty)} = \max(X, \langle \text{ACT} \rangle \langle b \rangle X)$ (not expressible in MTL) is violation monitorable.

Theorem

For all $\phi \in VT^{lin}$, $(\|\phi\|)$ is sound and strongly violation complete for ϕ .

VT^{lin} : another example

(Unconstrained) bounded response for any bounded I

$$\max(Y, ([b]ff \vee \zeta) \wedge \langle \text{ACT} \rangle Y))$$

with:

$$\zeta = x \text{ \textbf{in} } \min(X, (y \text{ \textbf{in} } \langle y - x \in I \rangle \langle c \rangle \text{tt}) \vee (\langle \text{ACT} \rangle X))$$

is violation monitorable.

ST^{lin} : the satisfaction-monitorable fragment of T^{lin}

- Intuition : “relax” the syntax of the complete-monitorable fragment to allow general **min** formulae.

$$\begin{array}{llll}
 \phi, \psi & ::= & \texttt{tt} & | \texttt{ff} & | \phi \vee \psi & | \phi \wedge \psi \\
 & | & x \underline{\text{in}} \phi & | \langle a \rangle \phi & | \langle g \rangle \phi \\
 & | & \min(X, \phi) & | X \\
 & | & \cancel{\max(X, \phi)} & | \cancel{x \underline{\text{in}} \langle g_x \rangle \langle a \rangle X}
 \end{array}$$

- example: $\phi = \min(X, (x \underline{\text{in}} \langle b \rangle y \underline{\text{in}} \langle y - x \in (0, 2] \rangle \langle c \rangle \texttt{tt}) \vee \langle \text{ACT} \rangle X)$
expressing the property:

“there exists a **b** immediately followed by a **c** within $(0, 2]$ ”
is satisfaction monitorable.

Theorem

For all $\phi \in ST^{lin}$, (ϕ) is sound and strongly satisfaction complete for ϕ .

ST^{lin} : another example

The following formula, expressing the property:

“there exists a **b** such that every action within $(1, 7)$, relative to the occurrence of such **b**, is a **c**.”

$$\min(Y, ((\langle b \rangle \mathbf{tt} \wedge \zeta) \vee \langle \text{ACT} \rangle Y))$$

with:

$$\zeta = x \mathbf{in} \max(X, (y \mathbf{in} [y - x \in (1, 7)] \langle c \rangle \mathbf{tt}) \wedge (\langle \text{ACT} \rangle X))$$

is satisfaction monitorable.

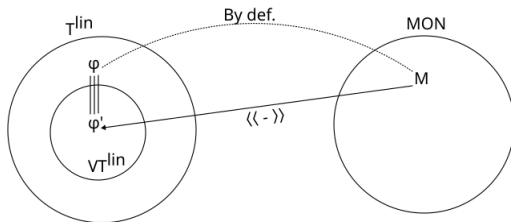
Bounded intervals and monitorability

Bounded interval does **not** imply monitorable

The MTL formula $\phi = G (c \implies F_{[2,5]} (b \wedge F c))$ is not monitorable.

- ▶ For violations, take $\pi = (c, 1)(b, 4)(b, 8)(b, 9)(b, 10) \dots$. π is not in $\llbracket \phi \rrbracket$.
- ▶ Yet, there is no way for a monitor to safely say NO based on a finite prefix of $\pi \dots$
 - ▶ ... otherwise such monitor would be unsound.
- ▶ For satisfactions, take $\pi' = (b, 1)(b, 2)(b, 3) \dots$. π' is in $\llbracket \phi \rrbracket$.
- ▶ Yet, there is no way for a monitor to safely say YES based on a finite prefix of $\pi \dots$
 - ▶ ... otherwise such monitor would be unsound.

Maximality of VT^{lin} and ST^{lin}



M sound and violation complete for ϕ and ϕ' implies $\llbracket \phi \rrbracket = \llbracket \phi' \rrbracket$.

Theorem

1. VT^{lin} is the largest violation-monitorable fragment of T^{lin}
2. ST^{lin} is the largest satisfaction-monitorable fragment of T^{lin}

Therefore $MT^{lin} = VT^{lin} \cup ST^{lin}$ is the largest monitorable fragment of T^{lin} .

Theorem

CT^{lin} is the largest complete-monitorable fragment of T^{lin} .

Conclusion: summary of contributions

This work is the first that:

- ▶ proposes a timed μ -calculus (T^{lin}) strictly more expressive than MTL
 - ▶ with complete proofs on monotonicity, continuity and expressiveness.
- ▶ solves the monitorability problem for such an expressive timed logic
 - ▶ with complete proofs using e.g., a precise characterisation of unique fixed points,
 - ▶ and fully automated synthesis from MTL to T^{lin} and from T^{lin} to monitors.

Conclusion: future work

- ▶ Timed automata as monitors
- ▶ Bounded-memory monitorability
- ▶ Tight monitorability
- ▶ Overhead-aware monitorability

Thank you.
Any questions?

Constrained min. and max.

Definition

$$\text{app}_\nu(X, \phi, 0) = \text{tt}$$

$$\text{app}_\nu(X, \phi, k+1) = \phi[\text{app}_\nu(X, \phi, k) / x]$$

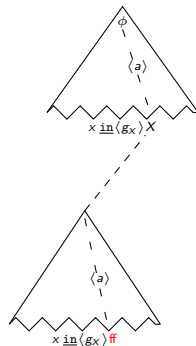
$$\text{app}_\mu(X, \phi, 0) = \text{ff}$$

$$\text{app}_\mu(X, \phi, k+1) = \phi[\text{app}_\mu(X, \phi, k) / x]$$

Lemma

For all $\phi \in CT^{lin}$,

- ▶ $\llbracket \max(X, \phi) \rrbracket_\nu = \bigcap_{n \in \mathbb{N}} \llbracket \text{app}_\nu(X, \phi, n) \rrbracket_\nu$
- ▶ $\llbracket \min(X, \phi) \rrbracket_\nu = \bigcup_{n \in \mathbb{N}} \llbracket \text{app}_\mu(X, \phi, n) \rrbracket_\nu$



Operational semantics of monitors

$$\text{MACT} \quad \frac{}{(a.M, v_c, v_t) \xrightarrow{a} (M, v_c, v_t)}$$

$$\text{MLET} \quad \frac{(M, v_c, v_t[x := v_c]) \xrightarrow{a} (M', v_c, v'_t)}{(\text{LET } x \text{ IN } M, v_c, v_t) \xrightarrow{a} (M', v_c, v'_t)}$$

$$\text{MSELL} \quad \frac{(M, v_c, v_t) \xrightarrow{a} (M', v_c, v'_t)}{(M + N, v_c, v_t) \xrightarrow{a} (M', v_c, v'_t)}$$

$$\text{MSELR} \quad \frac{(N, v_c, v_t) \xrightarrow{a} (N', v_c, v'_t)}{(M + N, v_c, v_t) \xrightarrow{a} (N', v_c, v'_t)}$$

$$\text{MVER} \quad \frac{M \in E}{(M, v_c, v_t) \xrightarrow{a} (M, v_c, v_t)}$$

$$\text{MTIME} \quad \frac{}{(M, v_c, v_t) \xrightarrow{d} (M, v_c + d, v_t)}$$

$$\text{MREC} \quad \frac{(M[\text{rec } X.M/X], v_c, v_t) \xrightarrow{a} (M', v_c, v'_t)}{(\text{rec } X.M, v_c, v_t) \xrightarrow{a} (M', v_c, v'_t)}$$

$$\text{MGUARD} \quad \frac{(M, v_c, v_t) \xrightarrow{a} (M', v_c, v'_t) \quad v_t \models g}{(g.M, v_c, v_t) \xrightarrow{a} (M', v_c, v'_t)}$$

$$\frac{M \odot N \notin E \quad (M, v_c, v_t) \xrightarrow{a} (M', v_c, v'_t) \quad (N, v_c, v_t) \xrightarrow{a} (N', v_c, v''_t)}{(M \odot N, v_c, v_t) \xrightarrow{a} (M' \odot N', v_c, v'_t \sqcup v''_t)}$$